

VAISHNAVA RUCHIKA PUPPALA

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PROFESSIONAL SUMMARY

Detail-oriented **Information Technology graduate** with expertise in **Python**, **SQL**, and **Machine Learning**. Experienced in **data preprocessing**, **automation**, and **real-time data processing** using **Python** to drive efficiency and optimization. Strong foundation in **MATLAB** for **signal processing** and **visualization**. Proficient in **object-oriented programming (OOP)** and collaborating with cross-functional teams to implement effective **technology solutions**. Proven ability to leverage **data analysis** skills to improve workflow efficiency and deliver actionable insights that drive business outcomes.

SKILL SUMMARY

- **Programming Languages:** Python, Java, SQL, MATLAB, Object-Oriented Programming (OOP)
- **Data Science:** Machine Learning(ML), Data Analysis, Sensor and Real-Time Data Processing, Image Processing, Visualization
- **Software & Tools:** Power BI, Tableau, Figma, Canva, Balsamiq Microsoft Office Suite (Excel, Word, PowerPoint, Outlook, Teams, Access, Visio, Project), NumPy, Pandas, Matplotlib, OpenCV, Photoshop
- **Other Expertise:** Internet of Things(IoT) Device Design, Debugging, Optimization, Technical Documentation, Problem-Solving, Project Management, UX/UI, Wireframing, Low and High-Fidelity Designs and Prototypes, Photography, Editing
- **Soft Skills:** Team Collaboration, Mentoring, Feedback & Progress Tracking, Teaching, Technical Communication, Research

WORK EXPERIENCE

FLORIDA STATE UNIVERSITY (On-site)

Florida

Teaching Assistant

August 2023 - December 2024

- Supported three undergraduate courses, including Networking and Telecommunication, Object-Oriented Programming, and Web Applications, optimizing course management and improving student engagement across various platforms.
- Managed assignments and assessments for 40 students per class, enhancing technical expertise in grading, feedback, and progress tracking while implementing structured methods to streamline course administration and evaluation.
- Conducted review sessions, addressed inquiries via Canva, and Zoom meetings, and tracked attendance, fostering collaboration and deepening students' understanding of programming concepts and networking fundamentals.

STRIKETRU (Remote)

India

Python Developer Intern

May 2024 - August 2024

- Developed Python scripts to automate data processing workflows and perform basic image analysis tasks, significantly improving overall workflow efficiency and reducing manual efforts across team operations.
- Assisted in identifying, debugging, and optimizing Python code to enhance performance, ensure functionality, and deliver reliable solutions seamlessly integrated into existing systems and processes.
- Conducted detailed data preprocessing activities such as cleaning, organizing, and transforming datasets while meticulously documenting workflows to improve project consistency, team collaboration, and productivity.

COGNIZANT (Remote)

India

Software Program Analyst Trainee Intern

May 2022 - November 2022

- Completed a rigorous 3-month training program focused on foundational programming skills in Python and Java, building a strong understanding of software development principles within a collaborative learning environment.
- Gained proficiency in Python and foundational Java, mastering object-oriented programming, core Java features, and essential frameworks, strengthening problem-solving skills for software development tasks.
- Engaged in hands-on coding exercises and collaborated on projects, benefiting from mentorship that enhanced analytical thinking, adaptability, and readiness for a fast-paced IT environment.

UNIVERSITY PROJECTS

SLEEP STAGE DETECTOR

August 2021 - May 2022

- Developed a machine learning model in MATLAB for sleep stage detection, achieving 85% accuracy, validated with 10 patient records. Integrated Python for data preprocessing and analysis, using MATLAB for signal processing and visualization, ensuring efficient project execution.

PARIETAL EYE

August 2021 - January 2022

- Designed an IoT device using Arduino for vehicle blind spot detection, processing sensor data for real-time communication. Leveraged Python for data optimization and seamless integration, achieving 90% accuracy in detecting blind spots and reducing collision risks by 25% which the department represented during the university expo.

SMART STICK

December 2019 - May 2020

- Developed an IoT-based assistive device for visually impaired users using Python and Arduino. Integrated ultrasonic sensors for real-time obstacle detection within a 2-meter range, providing haptic feedback and audio alerts. Achieved 95% accuracy to enhance navigation and user experience.

EDUCATION

FLORIDA STATE UNIVERSITY

Florida

Master of Science in Information Technology

January 2023 - December 2024

Cumulative GPA: **3.85/4.0**;

Relevant coursework: Fundamentals of Systems Engineering, Information Systems Management, Introduction to Information Architecture, Usability Analysis, Introduction to Project Management, Leadership in Technology, and Product Design.

B V RAJU INSTITUTE OF TECHNOLOGY

India

Bachelor of Technology in Information Technology

August 2018 - May 2022

Cumulative GPA: **3.1/4.0**;

Relevant coursework: Machine Learning, Database Management Systems, Data Structures, Design & Analysis of Algorithms, Object-Oriented Programming Through Java, Software Engineering, and Software Testing Methodologies.

CERTIFICATIONS

- Graduate Certificate in Information Leadership and Management (FSU)
- UX Design Professional Certificate (Google)
- Selected Coursera Certifications:
 - Python, Introduction to Computer Vision and Image Processing, HTML, SQL
 - OOP in Java, Exploratory Data Analysis With Python and Pandas, Foundations: Data, Data, Everywhere

ORGANIZATIONS

- College of Communication & Information (CCI) Student Leadership Council (SLC) – Member
 - Participated in the Homecoming Parade at Florida State University, contributing to event coordination and promotion.
- Indian Students Association of Tallahassee – Volunteer (Mar 2023)
 - Assisted with event planning and performed dance activities at organized cultural events.
- Promethean 2K19 – Volunteer (Oct 2019)
 - Participated in the "Think Quick" event, supporting IT department activities and event logistics.